RESEARCH ARTICLE SUMMARY

COGNITIVE MAPS

Hippocampal coding of identity, sex, hierarchy, and affiliation in a social group of wild fruit bats

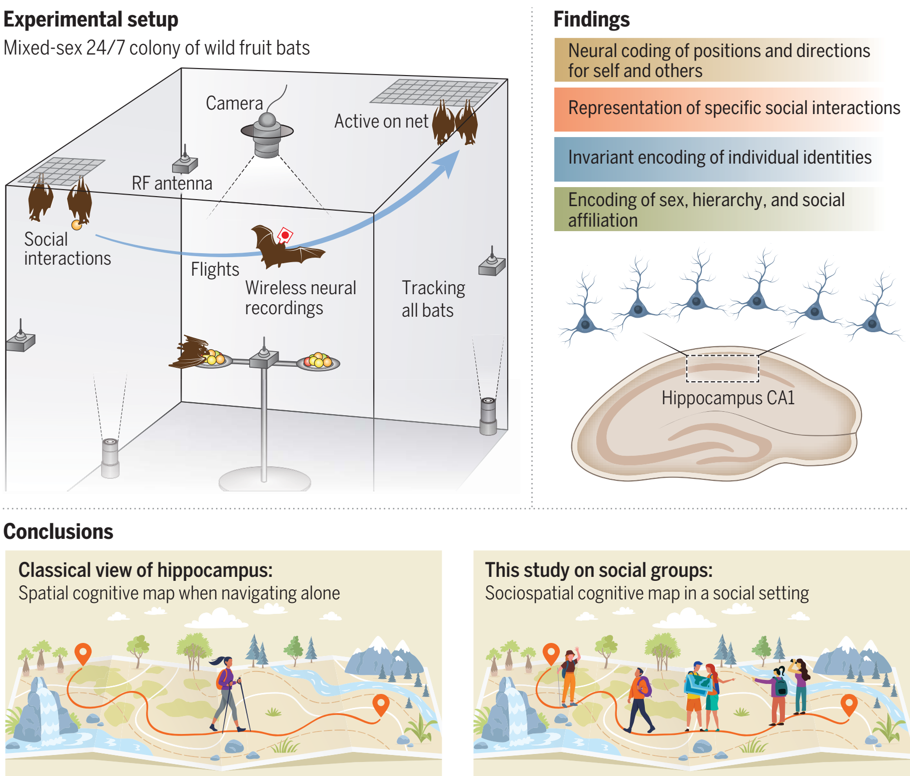
Saikat Ray, Itay Yona, Nadav Elami, Shaked Palgi, Kenneth W. Latimer, Bente Jacobsen, Menno P. Witter, Liora Las*, Nachum Ulanovsky*

INTRODUCTION: Social animals live in groups and interact volitionally in complex ways. To perform real-life social behaviors, the brain needs to code other individuals' identities, represent various types of social interactions, and encode key social factors such as the sex, dominance hierarchy, and social affiliation of multiple other individuals. However, our understanding of how the brain deals with such diverse requirements stems from constrained laboratory experiments in which an animal typically exhibits one specific behavior with one other animal in one particular task. This leaves the fundamental question unexplored: How does the brain actually represent the real world with its complex, multianimal settings?

RATIONALE: To understand natural social coding in the mammalian brain, we studied Egyptian fruit bats (*Rousettus aegyptiacus*), which are highly social mammals, and focused on the hippocampus, a brain area that in previous studies has been shown to be important for memories of social identities, episodic events, and spatial locations. We hypothesized that in natural scenarios, when all of these disparate aspects occur simultaneously, hippocampal neurons would bind together all of these different types of information. To create a naturalistic environment, we established a laboratory-based "cave," housing mixed-sex groups of five to 10 wild-caught bats. The bats lived together continuously (24/7) for several months, engaging in natural social behaviors without any imposed tasks. During this time, we conducted wireless neural recordings from the dorsal hippocampus area CA1 of both male and female bats and tracked their positions, head directions, and social interactions with each other.

RESULTS: The freely behaving bats formed a stable social network and displayed three key behaviors: (i) flying between two nets located at opposite corners of the setup, (ii) engaging in social interactions, and (iii) being active on the net and observing each other. We found that hippocampal "place cells," neurons known to represent the animal's own position, were modulated during flight by the social context, i.e., whether the bat was flying to meet another bat or to be alone. These cells also encoded the identities of other bats. This identity coding was invariant to the bat's flight direction. We also found that many hippocampal cells encoded social-interaction events, with different neurons typically encoding distinct types of social interactions such as affiliative grooming or aggressive boxing. During active observation on the nets, we used methods from machine learning and game theory to reveal that neurons encoded the bat's own position and head direction, together with the positions, directions, and identities of multiple other individuals. Identity-coding neurons encoded the same specific bat across different locations and different behavioral states, both in-flight and on the net, providing another example of social invariance. The strength of identity coding was modulated by the sex, dominance hierarchy, and social affiliation of the other bats.

CONCLUSION: Our use of a naturalistic social colony allowed us to discover that the classical hippocampal cognitive map of space also integrates rich social information, forming a sociospatial cognitive map. We found neurons that encoded social interaction events, identities and sex of other individuals, dominance hierarchy, and social affiliation, along with the position and direction of both self and others. These findings combine the historically disparate views on hippocampal function, which suggested that the hippocampus is important for encoding memory, social identity, or spatial maps. Here, we have shown that all of these factors are represented together in the same neural network. ■



Hippocampal place cells also encode social information, forming a sociospatial cognitive map. In a naturalistic mixed-sex colony of freely behaving wild Egyptian fruit bats, hippocampal CA1 neurons encoded social interactions with other bats and represented the identity, sex, dominance hierarchy, and social affiliation of other individuals. [Map illustrations: Juliana Brykova/Shutterstock]

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RESEARCH ARTICLE

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Hippocampal coding of identity, sex, hierarchy, and affiliation in a social group of wild fruit bats

Saikat Ray[1], Itay Yona[1], Nadav Elami[1], Shaked Palgi[1], Kenneth W. Latimer[2], Bente Jacobsen[1,3], Menno P. Witter[3], Liora Las[1]*, Nachum Ulanovsky[1]*

Social animals live in groups and interact volitionally in complex ways. However, little is known about neural responses under such natural conditions. Here, we investigated hippocampal CA1 neurons in a mixed-sex group of five to 10 freely behaving wild Egyptian fruit bats that lived continuously in a laboratory-based cave and formed a stable social network. In-flight, most hippocampal place cells were socially modulated and represented the identity and sex of conspecifics. Upon social interactions, neurons represented specific interaction types. During active observation, neurons encoded the bat's own position and head direction, together with the position, direction, and identity of multiple conspecifics. Identity-coding neurons encoded the same bat across contexts. The strength of identity coding was modulated by sex, hierarchy, and social affiliation. Thus, hippocampal neurons form a multidimensional sociospatial representation of the natural world.

From a hive of bees (*1*) to a troop of baboons (*2*) or a society of humans, social groups are ubiquitous in nature, and social interactions in a group are crucial for survival (*3*). However, neurophysiological investigations of social interactions are typically conducted in pairs of animals (*4*–*10*), not in a group. Furthermore, social group behavior in the natural world is complex. For example, one may choose to mate with one individual, fight with another, and play with a third, a widespread scenario in real life that is not captured by typical laboratory experiments in social neuroscience. Such real-life situations require the representation of individuals' identities and of features pertinent to it, such as the sex, social hierarchy, and social affiliation of multiple other individuals, and representing the social interactions themselves. To date, no studies have examined together the representation of identity, sex, hierarchy, social affiliation, and social interactions in the same neuronal population. Further, there have been no studies that performed neural recordings in a mixed-sex group or in a group of wild social animals—in any species and in any brain area. Here, we aimed to close these three major gaps to elucidate how the brain represents social information in rich multianimal settings like those that would be encountered in the natural world. We focused on the hippocampus, which was shown in recent years to be important for social memory (*4*, *11*–*13*). In particular, we studied its output structure, CA1, an area containing neurons that represent the animal's own position (*14*–*16*) as well as neurons representing other animals' positions in a pairwise social setting (*5*, *6*). We investigated highly social wild mammals, Egyptian fruit bats (*Rousettus aegyptiacus*) (*17*–*20*), and allowed them to socialize freely to understand how hippocampal neurons represent identity, sex, hierarchy, social affiliation, and social interactions in natural social groups.

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Continously monitoring a mixed-sex social group of wild bats

We established a laboratory-based cave for housing mixed-sex colonies of five to 10 wild-caught Egyptian fruit bats (Fig. 1A; $n = 5$ groups for the $n = 5$ recorded bats, with approximately equal proportions of males and females in each group). The composition of the group was stable over long periods of time (weeks to months) and was changed only when we replaced the recorded bats. Egyptian fruit bats are a long-lived, highly social species with complex social structures (*17*–*20*). The bats in our experiment were familiar with one another and lived in this setup continuously (24/7) for several months. They were free to fly and interact with each other and could choose between two nets to roost on (Fig. 1A; room dimensions: 2.7 × 2.3 × 2.6 m; large net, 40 × 80 cm; small net, 40 × 40 cm). We tracked the locations and identities of all the bats continuously during the recording session. During the flights, we tracked them in three dimensions (3D) using a radiofrequency-based system with a set of unique tags that the bats carried (10-cm precision) (*21*). We also placed barcodes on all of the bats (fig. S1) (*22*) and used a set of video cameras to track their identities, positions, and head directions at high resolution (millimeter-level precision) while they were on the nets. We also used video recordings to document in detail the bats' social interactions (movie 1). We performed wireless electrophysiological recordings from one or two bats for up to 3 hours per experimental day for a total of 99 recording days. These recording sessions commenced shortly after the beginning of the day's dark phase (Fig. 1B), when the bats are most active (*17*, *18*). No humans were present inside the room during the recordings.

A key aspect of natural social groups is stability in their social structure (*23*, *24*). Our bat colony exhibited a stable social network over weeks, with stable social-affiliation preferences (Fig. 1, C and D) (*21*). Another integral aspect of stable social groups is the stability of their social hierarchy (*25*, *26*). Indeed, our bat colony displayed a stable social hierarchy over many months (Fig. 1E and fig. S2). The social hierarchy was also transitive and linear (transitivity: 84%, $n = 113/134$ triads; linearity: $h = 0.80$, $P < 10^{-6}$; h ranges from 0 to 1, with 1 indicating perfect linear hierarchy and 0 indicating perfect equality) (*21*). Together, these features indicate that we successfully created a highly social, naturalistic, multianimal society and that bats could recognize each other—i.e., they formed internal representations of the identities of individual conspecifics within the group, which allowed them to maintain social preferences.

Social behavior in the natural world is unconstrained, so we did not constrain the behavior of our colony of bats to any particular task. Rather, the bats volitionally chose which behaviors to perform and when, where, and with which other bats. We focused our analysis on the three most prominent behavioral states that the bats exhibited: (i) flying between the two nets, (ii) engaging in social interactions on the nets, and (iii) being active on the nets and observing other bats (Fig. 1, F and G). The bats' activity on the nets in state iii was evident by their substantial head rotations and very low rate of sharp-wave ripples during that period (fig. S3). These three behavioral states occurred in an interleaved manner, with the bats spending the largest fraction of time on the large net and being active on it while intermittently engaging in social interactions and flights (Fig. 1G). While bats engaged in these three behaviors, we recorded 489 well-isolated neurons from the dorsal CA1 region of the hippocampus from five adult bats (three males and two females) (Fig. 1H). Of the 489 neurons, 444 cells met the inclusion criteria of having sufficient behavioral data and enough spikes to be valid for analysis in at least one of the three behaviors (see table S1 for the numbers of cells that were valid for analyzing each of the three types of behaviors) (*21*).

Hippocampal neurons exhibit social modulation and identity coding in flight

We first investigated how hippocampal cells respond in the flight state. The bats flew mostly

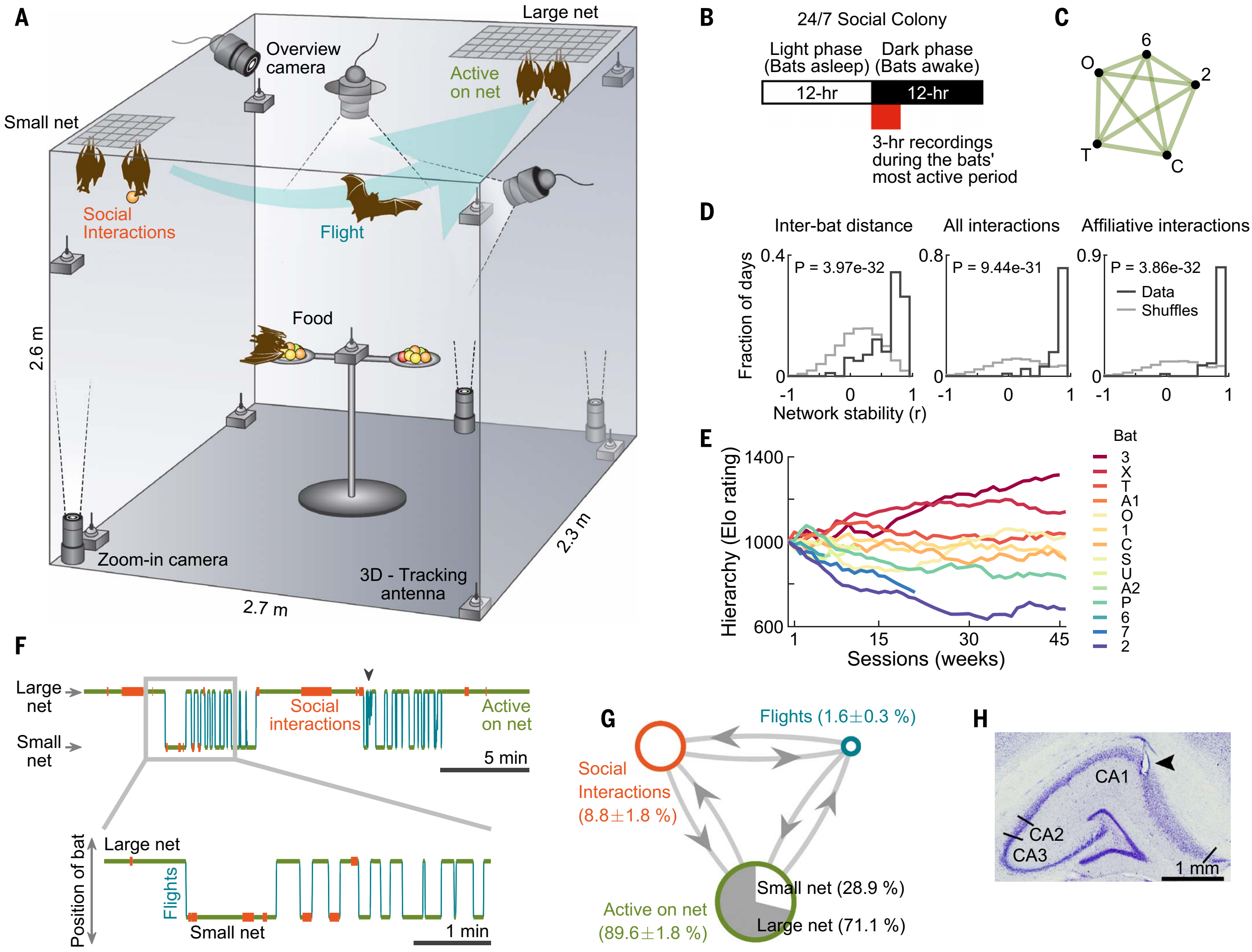

Fig. 1. Setup and behavior: A stable social network in a colony of freely behaving bats. (**A**) Five to 10 male and female bats lived continuously in a room (2.7 × 2.3 × 2.6 m) simulating a bat cave. Their social interactions and activity on two nets were observed using video cameras, and their flights were tracked by a radiofrequency-based 3D tracking system. (**B**) Schematic illustration of the daily timeline in the colony. (**C**) The average social network for all days in which this group of five bats were together. The network was constructed based on the physical distances between all pairs of bats, which is a proxy for social affiliation (n = 47 days; the colony's social composition was the same throughout these 47 days) (*21*). Nodes of the graph correspond to individual bats (bats C and T are males; bats O, 6, and 2 are females; bat 6 is the recorded bat), and distances between nodes are proportional to the median distances between the bats. (**D**) Distributions of social network stability quantified by correlation (Pearson's r) between the vector of graph centralities for each recording day and the vector of median graph centralities of that social network across all days [n = 99 days in total; n = 5 different networks (groups) for the five recorded bats] (*21*). Network stability was computed based on distances between all pairs of bats (left), rate of all social interactions (middle), or rate of affiliative social interactions with the recorded bat (right). Black line is the data; gray line is the shuffled network (*21*). Shown are P values for Wilcoxon rank-sum test between distributions of data versus shuffles; note the high network stability for the data compared with shuffles. (**E**) Dominance hierarchy (Elo rating) of individual bats over time (weeks). Each curve corresponds to an individual bat (n = 14 bats that participated in our five groups; bats 3, O, 1, 6, 7, and 2 are females, and bats X, T, A1, C, S, U, A2, and P are males). The hierarchy was determined in a separate setup using a pairwise hierarchy test (fig. S2A) (*21*). Hierarchy testing for some bats started later than others; the Elo ratings were determined in the original timing of the hierarchy tests, and then the curves for individual bats were shifted to their first session for display purposes. (**F**) Example ethogram of a recorded bat (top, 25 min from a typical session of 2 to 3 hours; bottom inset, zoom-in on 5 min). Shown are three interleaved behavioral states during the session: flights (blue), social interactions of the recorded bat with other bats (red), and activity on the two nets (green). Black arrowhead at top indicates an instance of irregular flights. (**G**) Summary of time spent by all the recorded bats in the three behavioral states and transitions between the states. Blue indicates time spent flying; red, engaging in social interactions; and green, being active on the nets. The bats spent most of their time being active on the large net, intermittently engaging in social interactions and flying. (**H**) Coronal section through dorsal hippocampus of one recorded bat; arrowhead, electrolytic lesion in CA1 at the end of a tetrode track.

in direct flights between the two nets, and we only used these direct flights for analysis. We analyzed the two flight directions separately, as is commonly done in studies of hippocampal place cells (*27*, *28*). We found that 61% of the cells were significant place cells in-flight (n = 149/243 cells × directions; significance was assessed by comparing them with spike shuffles; 243 cells × directions contained enough behavioral data and enough spikes to analyze place tuning in-flight) (*21*). This fraction of 61% place cells is consistent with previous reports on bats flying in directed trajectories in a medium-sized room (6, 28). We then investigated whether place tuning during flight depends on social context. For example, do neurons represent locations similarly when a bat flies to "meet" another bat and when it flies to be alone? We were able to investigate this question because of the natural asymmetry in the social preference of the two nets, with bats being only intermittently present on the small net (Fig. 1, F and G). We found that hippocampal cells were highly affected by social context (Fig. 2, A to D). Firing rate (FR) maps were strongly modulated in-flight between the condition when the bat was landing alone on the net or when it was landing while other bats were present on the net [Fig. 2, A and B, "landing," compare social maps (right) with alone maps (left)]. In addition to prospective social modulation, which depended on where the bat was going (landing), we also found retrospective social modulation, which depended on where it came from, i.e., whether the bat took off from a net when it was alone or if the net had other bats on it (Fig. 2, C and D, "takeoff"). The firing rate of socially modulated cells could be up-modulated or down-modulated in the social condition compared with the alone condition (Fig. 2, A to D), with simultaneously recorded cells showing heterogeneous responses (e.g., Fig. 2B, cells 70 and 71).

We defined significant socially modulated cells based on a bootstrap shuffle analysis in which individual flights were randomly redistributed between the social and alone conditions. A cell was considered significant if the root-mean-square difference (RMSD) in the 2D firing-rate maps between the social and alone conditions was greater than or equal to the 95th percentile of the shuffles (Bonferroni corrected for the two tests for landing and takeoff) (21). A majority of hippocampal cells (67%) were significantly socially modulated in-flight in at least one flight direction (Fig. 2E, "all cells," left bar; 67 of the 100 cells that were valid for analysis, i.e., with sufficient behavioral data in both social and alone conditions; see fig. S4D for counterexamples of cells that did not show social modulation) (21). We found similar proportions of cells that were socially modulated by bat presence or absence at the landing net (55 of 67) and at the takeoff net (49 of 67); 37 cells were modulated by both the landing net and the takeoff net.

To control for effects of spatial coverage and flight kinematics, we took several steps. (i) Comparisons between maps were only made on the overlapping regions of the maps (fig. S4, A to C). (ii) The landing and takeoff locations and 3D flight trajectories between the social and alone conditions were highly overlapping (fig. S5, A and B). (iii) Social coding was similar for the subset of social flights with the most similar trajectories or the most different trajectories to the alone flights (fig. S6C). (iv) Social modulation of neurons was not correlated with trajectory variability (fig. S6, E and F) (21), thus ruling out the possibility that the neural-activity differences between social and alone flights might arise from differing trajectories (29). (v) Flight velocity, acceleration, and flight curvature (straightness index) were very similar between the social and alone conditions (fig. S5, C and D), indicating that kinematic differences could not underlie the neural-activity differences between social and alone flights. (vi) Finally, we also controlled for the different number of flights in the social and alone conditions by matching the numbers of flights, and did not find substantial changes in the fraction of social-coding cells (fig. S6B).

Significant socially modulated place fields were found throughout the flight trajectory (Fig. 2F; n = 82 cells × directions). There was a slight overrepresentation of social modulation near the landing and takeoff locations (Fig. 2G), which was similar to the general distribution of place fields, as well as to the distribution on nonsocially modulated place fields (fig. S5, E and F; Kolmogorov-Smirnov tests comparing Fig. 2G with fig. S5E, $P = 0.79$ and $P = 0.64$ for the two directions; comparing Fig. 2G with fig. S5F: $P = 0.54$ and $P = 0.59$).

The socially modulated neurons showed a large effect size: The mean social modulation depth was twofold compared with nonsocial cells (Fig. 2H, compare the black and gray lines; Wilcoxon rank-sum test: $P = 2.94 \times 10^{-26}$; see also fig. S7A) (21), and the mean ratio of peak firing rates between the social and alone maps was 2.4-fold (Fig. 2I, mean of the black line). The social modulation was best explained by "rate remapping" of cells, which is a change in peak firing rate, rather than by "global remapping," which is a complete change in the map pattern (fig. S7B, note the high Pearson correlations between the two maps). The rate remapping was particularly prominent for cells showing large social modulation (fig. S7C, red histogram).

We observed a distinct behavior in which a bat would land on top of another bat instead of near it (Fig. 2J, right). This was likely an affiliative-landing behavior because it was typically followed by affiliative interactions such as allogrooming rather than by aggressive interactions. We then investigated whether neurons were modulated by this affiliative-landing behavior. Although this event was too rare on most days to allow systematic analysis, there were a small number of days when it occurred often enough to analyze (21). On these days, we found cells that were significantly modulated in-flight by the affiliative-landing behavior. The representative cell shown in Fig. 2J almost did not respond when the bat landed alone, responded moderately but significantly higher when it landed near other bats [nearby versus alone: effect size (social modulation depth) = 3.9, $P < 0.001$], and responded strongly when the bat landed on top of other bats (on top versus alone: effect size = 8.6, $P < 0.001$; on top versus near: effect size = 5.4, $P < 0.001$). Overall, of nine cells that had enough data for analysis and no velocity differences between the three conditions, we found four cells that were significantly modulated by affiliative landing. This further indicates a social modulation of place tuning in-flight.

Does the social modulation of cells depend on the identity of the conspecific involved (Fig. 3A)? To answer this, we analyzed a subset of the socially modulated cells that had enough data to compare social modulation for at least two other bats (Fig. 3, B to D; n = 84 cells). Many of these cells were identity coding and exhibited a significant difference between the social and alone conditions only for a particular conspecific bat, but not for others (Fig. 3B: examples; n = 34/84 cells) (21). The percentage of identity-coding cells was higher among the neurons that were nonplace cells (53%) compared with place cells (37%) (Fig. 3D). We also found a smaller subset of cells that were identity generalizing and exhibited a significant difference in firing when a bat was present or absent regardless of the bat's identity (Fig. 3C: examples; n = 15/84 cells). More than twice as many cells were significantly identity coding in-flight than identity generalizing (Fig. 3D, all cells: 40.5% identity-coding cells, 34/84; 17.9% identity-generalizing cells, 15/84) (21).

We next investigated whether the CA1 neuronal population could be used to decode the social context of the flight (social versus alone) and also determine whether a specific bat was present or absent (decoding of identity). We performed Bayesian maximum-likelihood decoding using the whole cell population (separately for the two flight directions) (21). We found that, first, we could indeed decode well above chance whether individual flights belonged to social or alone conditions (Fig. 3E, black error bar; mean decoding accuracy = 85%; t test for comparison with the chance level of 50%: $P = 0.0068$). Second, we were able to decode well above chance whether a specific individual bat was present or absent during a flight (Fig. 3E, yellow error bars; mean decoding accuracy for identity = 82%; t test for comparison with the chance level of 50%, pooling all bat identities together: $P = 0.0066$; t tests for individual bats: $P = 0.024$, $P = 0.018$, $P = 0.005$, and $P = 0.0001$). The fact that we could reliably decode bat identity further supports our finding that individual identity is strongly encoded by hippocampal neurons.

We next investigated whether the social modulation to specific individual bats was invariant to the flight direction. We found that, indeed, cells that were up-modulated by the presence (or absence) of a specific individual bat in one flight direction were also up-modulated

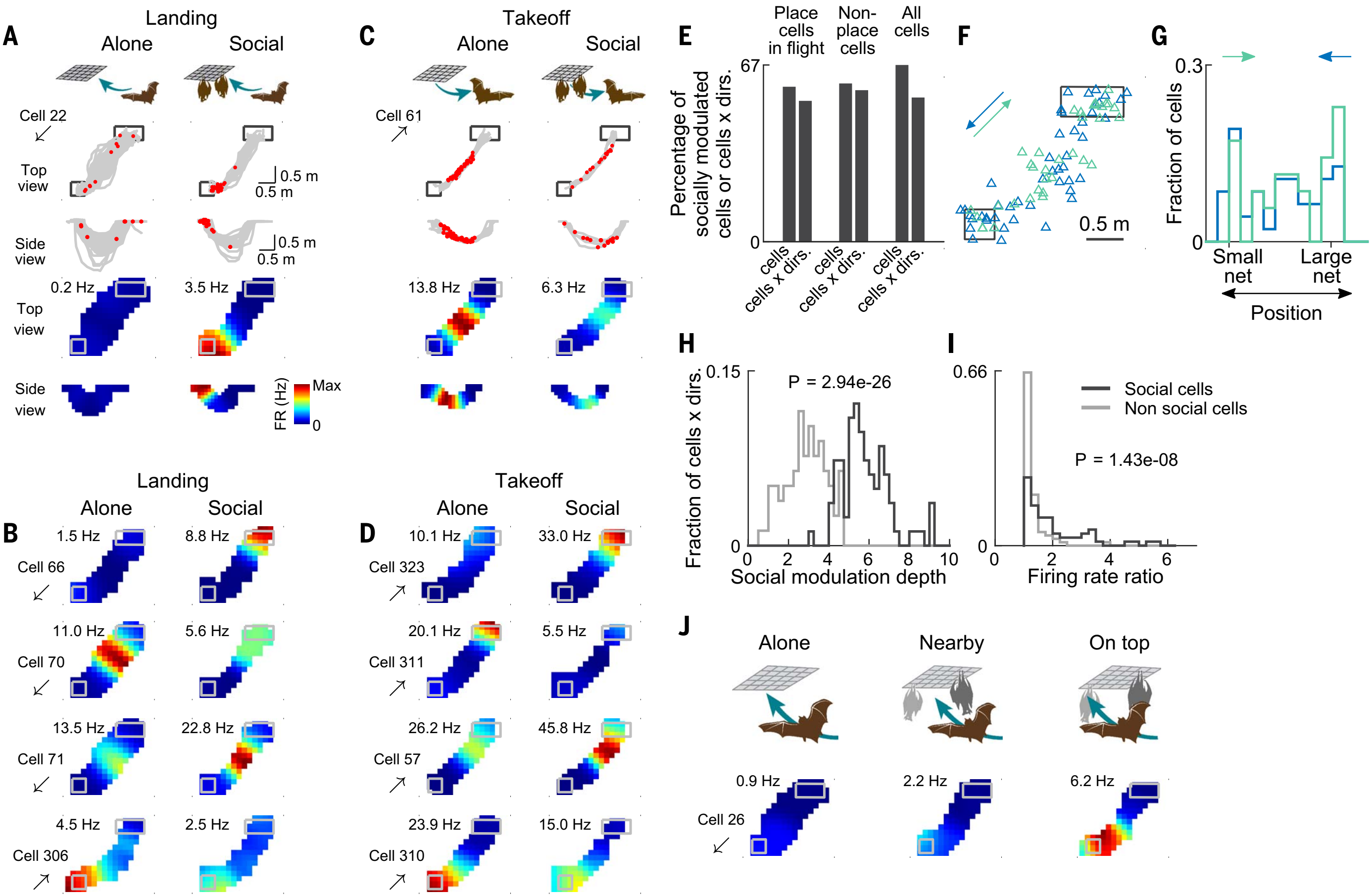

Fig. 2. Hippocampal place coding during flight is strongly modulated by social context. (**A**) Example of a socially modulated neuron during landing. The recorded bat landed on an empty net (left column, "alone") or on a net where other bats were present (right column, "social"). Second and third rows: flight trajectories (gray) with spikes overlaid (red), shown from top view (second row) and side view (third row). Fourth and fifth rows: 2D firing-rate (FR) maps corresponding to the raw data above. The maps are colored from zero (blue) to maximal firing rate [red; value indicated; each pair of maps (social and alone) has the same color scale; see color bar]. Arrow below the cell number indicates flight direction (here, from large net to small net). Note that the neuron barely fired when the bat landed alone (left), but had a clear place field when it landed with other bats present (right; $P < 0.001$) (21). (**B**) Examples of four neurons that were significantly socially modulated during landing. Shown are top-view 2D firing-rate maps of different neurons (rows) for the alone condition (left column) and the social condition (right column). Note that the socially modulated place fields occurred throughout the flight trajectory, and that a neuron could be either up-modulated in the social condition (cells 66 and 71) or down-modulated (cells 70 and 306). (**C**) Example of a neuron that was socially modulated by the presence or absence of bats during takeoff, namely whether the bat took off from an empty net (left, "alone") or from a net where other bats were present (right, "social") [plotted as in (A)]. This neuron had a higher firing rate when the bat took off in the alone condition ($P = 0.002$). (**D**) Examples of four neurons that were socially modulated by takeoff [plotted as in (B)]. (**E**) Percentage of significantly socially modulated neurons in-flight for place cells (left pair of bars), nonplace cells (middle pair of bars), and all cells (right pair of bars). Note that most hippocampal neurons were socially modulated [place cells, $n = 47/80$; place cells × directions, $n = 55/103$; nonplace cells, $n = 12/20$; nonplace cells × directions, $n = 27/47$; all cells, $n = 67/100$; all cells × directions, $n = 82/150$; numbers in the denominator indicate cells with sufficient behavioral data (21)]. (**F**) Population summary: Top view of the room showing the locations of peak firing of all the significantly socially modulated neurons (triangles) for flights from large net to small net (blue) or in the opposite direction (green) ($n = 82$ cells × directions, which include all of the significantly socially modulated cells × directions regardless of whether they were significant place cells). Note that these fields spanned all locations along the flight trajectories. (**G**) Distribution of the locations of peak firing of socially modulated neurons for flights in each direction. Position was projected onto the diagonal line connecting the two nets for the same cells as in (F). (**H**) Histograms of social modulation depth (effect size, defined as the RMSD between social and alone maps normalized by the SD of the RMSD for the shuffles) for the significant socially modulated cells (black, $n = 82$ cells × directions) and the nonsocial cells (gray, $n = 78$ cells × directions; Wilcoxon rank-sum test of black versus gray, $P = 2.94 \times 10^{-26}$) (21). (**I**) Histograms of ratios of peak firing rates in social and alone conditions (FRsocial/FRalone or FRalone/FRsocial, whichever is higher) in socially modulated cells (black) and nonsocial cells (gray; Wilcoxon rank-sum test, $P = 1.43 \times 10^{-8}$; x-axis has been clipped [zoom] for visualization only). (**J**) Example neuron that responded weakly when landing alone (left), moderately when landing near other bats (middle), and strongly when landing on top of other bats (right).

by the presence (or absence) of the same bat in the other flight direction (Fig. 3F: examples; Fig. 3G: population; Pearson $r = 0.55$, $P = 5.77 \times 10^{-6}$; no such correlation was found for the nonsocial cells: fig. S7D, Pearson $r = -0.06$, $P = 0.53$). Thus, unlike spatial coding in CA1, which is known to show remapping between two flight directions (28, 30) (as seen also in Fig. 3F), social identity coding showed invariance between the two flight directions.

Finally, we found that in-flight, identity coding was sex dependent, with a higher percentage of identity-coding cells representing males compared with females (Fig. 3H; χ^2 test: $P = 0.022$; notably, there was no significant difference if the sex of the identity-coded bat was the same or different from the recorded bat: χ^2 test: $P = 0.122$).

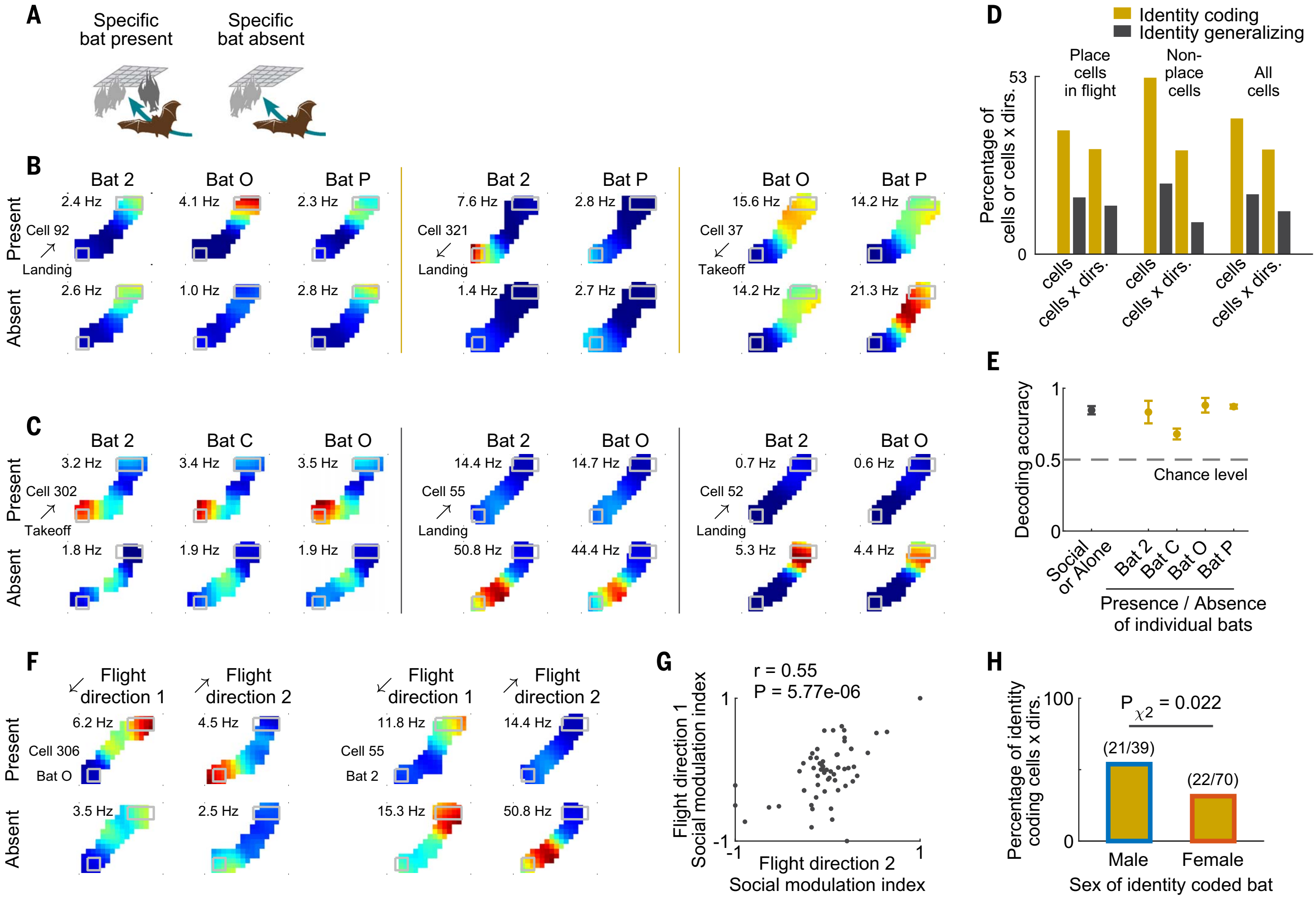

Fig. 3. Hippocampal neurons encode the individual identity of other bats. (**A**) Schematic illustration of flights when a particular conspecific bat is present (left) or absent (right). (**B**) Three example neurons that showed in-flight identity coding for specific bats. Shown are firing-rate maps computed for flights when a particular conspecific was present (top) or absent (bottom) for different conspecifics (columns). Left, neuron showing identity coding for bat O: Significant social modulation was seen when bat O was present compared with absent, but no social modulation is seen for the other two bats (bats 2 and P). Middle, neuron showing identity coding for bat 2: Significant social modulation is seen when bat 2 was present, but not for bat P. Right, neuron showing identity coding for bat P. (**C**) Three example neurons that generalized across bat identities [plotted as in (B)]. All three neurons showed significant social responses, with clear differences between when a bat was present or absent (significant differences between rows), but these responses were similar regardless of the identity of the bat (no differences between columns, i.e., no identity coding). (**D**) Percentage of identity-coding cells [yellow; cells as in (B)] and identity-generalizing cells [gray; cells as in (C)] showing that identity-coding cells were more abundant (identity coding: place cells, n = 24/65, place cells × directions, n = 26/83, nonplace cells, n = 10/19, nonplace cells × directions, n = 13/42, all cells, n = 34/84, all cells × directions, n = 39/125; identity-generalizing: place cells, n = 11/65, place cells × directions, n = 12/83, non-place cells, n = 4/19, nonplace cells × directions, n = 4/42, all cells, n = 15/84, all cells × directions, n = 16/125). Identity-coding and identity-generalizing cells could only be assessed when there were at least two other bats with enough trials for which comparisons could be made. Therefore, the numbers of identity-coding and identity-generalizing cells do not sum up to the total number of socially modulated neurons in-flight (Fig. 2E). (**E**) Decoding accuracy determined using a Bayesian maximum-likelihood decoder. We decoded whether a flight occurred in the social or alone conditions (black error bar; t test, P = 0.0068 compared with chance). We also decoded the presence or absence of specific individual bats during the flight (identity: yellow error bars; t-tests: P = 0.024, P = 0.018, P = 0.005, and P = 0.0001, for bats 2, C, O and P, respectively). Error bars indicate mean ± SEM (n = 4 conditions: the decoding was done separately for the 2 flight directions × 2 social responses to landing/takeoff). (**F**) Two example neurons showing social invariance to flight direction: a similar social modulation in both flight directions for the presence or absence of a specific individual bat. Shown are firing-rate maps computed for flights when bat O (left neuron) or bat 2 (right neuron) was present (top) or absent (bottom) for the two flight directions (columns). Left, a neuron showing social invariance to the presence of bat O on the small net, with higher firing when the recorded bat was flying toward the small net (flight direction 1) and also when it was flying away from the small net (flight direction 2). Right, a neuron showing social invariance to the absence of bat 2 on the large net (higher firing for the absence of bat 2 in both flight directions). (**G**) Population scatter of the social modulation index [contrast index of the peak FR of the maps when a specific individual bat was present versus absent $(FR_{present} - FR_{absent})/(FR_{present} + FR_{absent})$] for one flight direction versus the other flight direction. The social modulation index was significantly correlated between the two flight directions [Pearson r = 0.55, P = 5.77×10^{-6}; n = 59 cells × bats (including both identity-coding and identity-generalizing cells)]. We compared both directions for the same net being occupied by the individual other bat and included all neurons with enough data in both directions and in which at least one of the directions was significantly socially modulated. This positive correlation demonstrates invariant social modulation by individual bats, i.e., social invariance to flight direction, even though the spatial coding exhibited remapping between the two flight directions (F). (**H**) Left, percentage of identity-coding cells × directions representing male bats out of the potential number of male bats that could be assessed in the colony (males with enough data); right, similarly for female bats (χ^2 test for males versus females, P = 0.022; the χ^2 test considers the number of males and females that were encoded by the neurons out of the number that could be assessed).

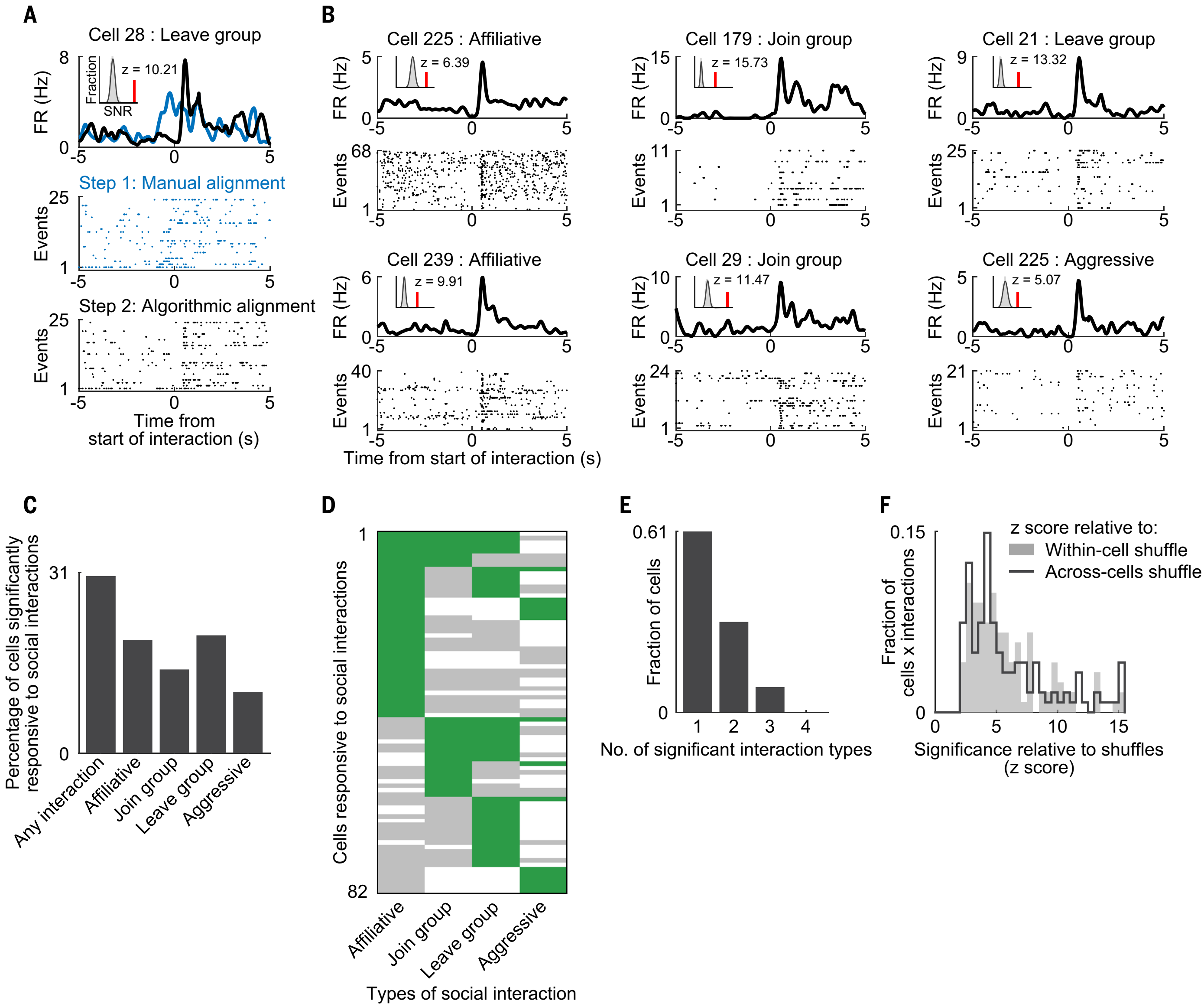

Fig. 4. Hippocampal cells respond to specific social interactions. (**A**) Example cell. Top, peristimulus time histograms (PSTHs) showing the average response (firing rate [FR] versus time) of a neuron to social-interaction events involving the recorded bat, for the social interaction type of leaving a group of bats. Bottom, spike rasters. Events (trials) are aligned to the start of interaction. Blue PSTH and blue raster: first step of interactions alignment based on manually annotated ethograms of the videos. Black PSTH and black raster: second step of interactions alignment, with interactions aligned to the model-estimated interaction time using the neural data itself (*21*, *31*). Inset in top panel shows the SNR for the actual PSTH (red line) and for the PSTHs of the within-cell shuffles (gray, histogram; black line, Gaussian fit; *z*-score of red line compared with Gaussian fit is indicated) (*21*). (**B**) Six additional example cells that were significantly modulated by different types of social interaction events: affiliative (friendly), joining a group, leaving a group, and aggressive interactions. Data are aligned as the black raster in (A); see movie 1 for examples of the different social interactions. (**C**) Percentage of cells significantly responsive to the various social interaction types, normalized separately for each interaction type by the number of cells valid for analysis. "Any interaction" indicates cells responsive to at least one interaction type. (**D**) Population summary of responses of hippocampal neurons (rows) to the various social interactions (columns). Green indicates a significant social response of a neuron to a particular interaction type; gray, no significant social response; white, could not assess the response due to a lack of the minimum number of interactions in that condition. (**E**) Selectivity to social interactions. Bar graph shows how many neurons responded significantly to only one, two, three, or all four social interaction types. Of the 82 neurons that responded significantly to social interactions, most (61%, 50/82 cells, left bar) only responded to a single type of social interaction. (**F**) Histogram of *z*-scores of the neuronal response (SNR) compared with shuffles for significantly responsive cells × interactions (n = 121 cells × interactions; gray bars, *z*-score compared with within-cell shuffles; black line, *z*-score compared with across-cells shuffles) (*21*). Medians of the black and gray histograms: z = 5.0 and z = 4.6, respectively.

Many hippocampal neurons respond to specific types of social interactions

We next examined the responses of dorsal CA1 neurons in the social-interactions behavioral state, i.e., when the recorded bat was involved in explicit social interactions. We categorized four commonly occurring interaction types: affiliative (friendly), joining a group, leaving a group, and aggressive interactions (see examples in movie 1). To align the start of the social-interaction event from an animal's perspective, which may be very different from the sequence of movements observable to us in the videos, we used two steps. In step 1, we determined the approximate start time of the

Movie 1. Social interactions in a colony of wild fruit bats. The movie shows examples of the social interactions that were most commonly observed in the colony of wild Egyptian fruit bats in our experimental setup. The bats most commonly engaged in affiliative interactions (allogrooming), joining a group, leaving a group, and aggressive interactions (boxing). These are the four types of interactions that were analyzed in Fig. 4. Shown are two example instances for each of these four types of social interactions: one example for bats before surgery and one example for implanted bats after surgery.

interaction event based on blinded manual annotation. In step 2, we used a method that estimates more precisely the time of firing-rate change in each social-interaction event from the neural activity itself (*21*, *31*) (Fig. 4A). Hippocampal cells showed significant responses to the four different types of social interactions (Fig. 4, A and B). These social-interaction responses could not be accounted for by velocity differences or by motor movements (fig. S8). In particular, only a minority of the cells responded to self-grooming compared with allogrooming (fig. S8B). Further, the social-interaction responses could not be explained as generalized responses to salient events because the cells did not respond to the nearby landing or takeoff of other bats, a highly salient type of event (fig. S8C).

Many hippocampal cells (30.4%) were significantly responsive to at least one type of social interaction (Fig. 4C, left bar; 82/270 cells; we analyzed here all the 270 cells with enough social-interaction events for analyzing social responses) (*21*). A cell's neural response to an interaction type was considered significant if the signal-to-noise ratio (SNR) of the response was large (≥2.5) and was greater than the 99th percentile of the SNR for two sets of shuffled data: within-cell shuffle and across-cells shuffle (*21*). Most cells were selective to particular social interactions, with 61% of the cells being selective to a single type of social interaction (fig. S9: examples; Fig. 4, D and E: population). The cells showed strong responses: Among the significant social-interaction cells, we found a mean z-scored response of $z = 5.9$ compared with within-cell shuffles and $z = 6.1$ compared with across-cells shuffles (Fig. 4F). These responses could not be explained by location-specific responses (place cells) because almost all of the cells (91.5%) were selective to only one or two types of social interactions (Fig. 4E) even though all four types of social interactions occurred at very similar locations (fig. S10A). For cells that responded to group joining, there was also a tendency to be significantly socially modulated in-flight during landing (χ^2 test: $P = 0.051$). This indicates that the same neurons had a tendency to encode the same social action, joining other bats, invariantly across behavioral states—regardless of whether the joining action was performed through flight or by moving on the net. Together, these results suggest that hippocampal neurons exhibit strong and selective responses to specific types of social interactions.

Encoding the position, direction, and identity of multiple conspecifics

Next, we investigated what hippocampal neurons represented while the bat was active on the net. We reasoned that a neuron could represent several sociospatial behavioral variables during this state: self-location or self-head direction, location of a specific other bat, or the direction and distance to a specific other bat (Fig. 5A). Because there were multiple conspecifics in the colony, this quickly became a highly multidimensional problem. To solve this computational challenge, we took a model-based approach in which we used the aforementioned behavioral variables to predict the responses of single neurons (Fig. 5B and fig. S11A). Specifically, we used a generalized additive model (GAM) (*32*, *33*) and trained two models independently for each cell, one representing allocentric variables [self-place, self-head direction, and the places of other bats (social place cells)] and another representing egocentric variables (directions and distances to other bats). The positions of the different bats were not correlated with each other and likewise the distances and directions to the different bats were not correlated with each other, allowing us to reliably assess each model separately (*21*). However the allocentric and egocentric variables were correlated with each other, precluding a reliable analysis of a mixed allocentric-egocentric model (*21*). We limited this analysis to include only data when at least three other bats were present on the net to allow comparing responses to different individual bats. Results did not change when we removed data around interactions and around sharp-wave ripples (fig. S11, B and C; the rate of ripples was very low, with median rate of 0.13 ripples/min, indicating that the bats were overall very active on the net). The models were fit with 10-fold cross-validation, and we further assessed whether the correlation between the neuronal activity in the data versus the model was significantly higher than the correlations computed for models fit to shuffled data [95th percentile of shuffles (rigid circular shuffling of spike trains)] (*21*). In cases in which both the allocentric and egocentric models provided significant fits, we performed an additional model-selection step using the deviance information criterion (DIC), a standard criterion for model selection (*31*), to determine which model (allocentric or egocentric) better explained the firing of the neuron (fig. S12A; the DIC analysis yielded very conclusive model selection) (*21*). After model selection, we proceeded to evaluate the individual contributions of each of the sociospatial behavioral variables in explaining the responses of the neuron. To this end, we fitted the model using all possible combinations of behavioral variables, and then used a game-theoretic approach of Shapley values (*34*, *35*) to evaluate the contribution of each behavioral variable (*21*). Shapley values could be reliably assessed because within each model, the behavioral variables were not strongly correlated (*21*).

We found that the responses of some cells were better explained by allocentric behavioral variables (Fig. 5, C to E) and some by egocentric behavioral variables (Fig. 5, F to H). Both sets of cells showed good correspondence between the data and the predictions of the model. High correlation of the cell's temporal firing-rate profile was found in the model versus the data [Fig. 5, C and F, top: compare green traces (model) with black traces (data); Spearman ρ = 0.54 and 0.58; these ρ-values are shown also as red lines in the inset (gray histograms are shuffles)]. Also, good correspondence was found between the firing-rate maps of the behavioral variables and the same maps predicted by the model (Fig. 5, C to H). Almost half of the hippocampal neurons were significantly explained by the model (44%, 172/394 cells). We additionally assessed an alternative egocentric model that also included the self-position and self-head direction variables and found that very few neurons were explained by this alternative model (fig. S11D, rightmost bar; 4% of the cells, 17/394). Thus, most egocentric cells tracked only other bats. Because of the small fraction of extra neurons that were captured by this alternative model, we used only the original egocentric and allocentric models (Fig. 5B).

Consistent with previous hippocampal studies reporting strong allocentric representations (*15*, *36*), we found here a larger fraction of allocentric cells than egocentric cells (Fig. 6A: 25% allocentric cells, 99/394; 19% egocentric cells, 73/394). These fractions were similar to previous reports on allocentric place cells and egocentric goal cells in bats navigating toward a goal (*37*). Many allocentric cells jointly encoded behavioral variables representing self and others (Fig. 6B, left matrix). Some of the cells encoded only a few behavioral variables, a sparse representation manifested by relatively

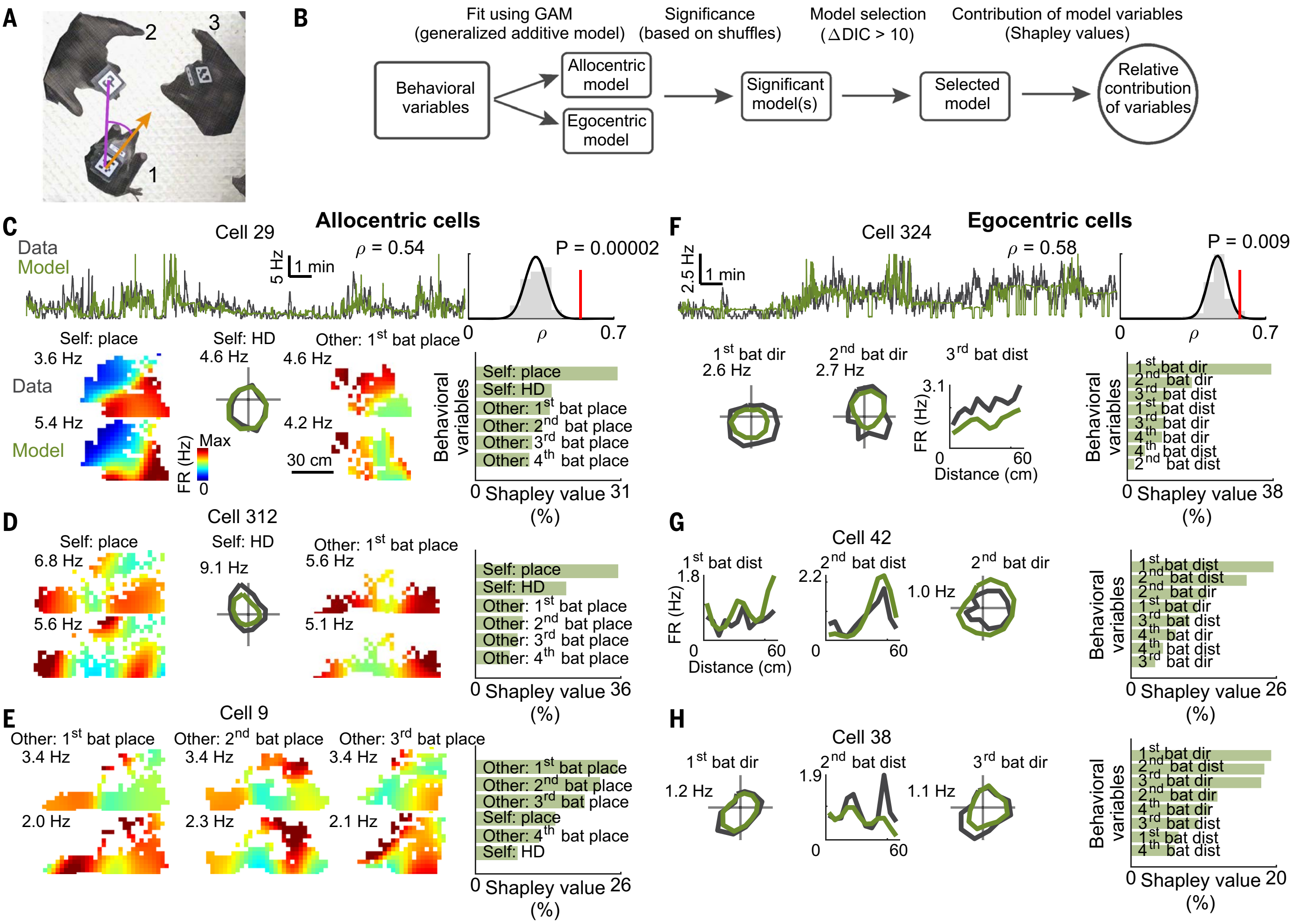

Fig. 5. Hippocampal neurons simultaneously represent position, direction, and distance for self and others. (**A**) Illustration of the behavioral variables for which neuronal coding was tested: self-position and self-head direction for the recorded bat (bat 1, orange), positions of other bats (bats 2 and 3), and directions and distances to other bats (e.g., to bat 2, purple). (**B**) Schematic illustration of the process for assessing whether a neuron represents allocentric or egocentric information and the contribution of the behavioral variables to the firing of the neuron (for an expanded schematic, see fig. S11A) (*21*). (**C**) Example allocentric cell. Top left traces, firing rate of the neuron over 20 min (black) and firing rate predicted by the GAM (green); note the high correlation of model to data (Spearman ρ = 0.54, computed over the entire session). Top right, correlation between data and model (red line) is significantly higher than correlations computed in a similar manner for models fitted to circularly shuffled spikes (gray, histogram for shuffles; black line, Gaussian fit) (*21*). Bottom right, relative contributions of the behavioral variables (Shapley values) to explaining the firing of this neuron. Three bottom left panels show tunings for the three most important variables (i.e., those with the highest Shapley values). Left, self-place firing-rate maps based on data (top) and based on model (bottom); note the similarity of the two maps. Middle, self head-direction tuning curves for data (black) and model (green); note their similarity. Right, tuning to the position of another bat based on data (top) and model (bottom). The tuning curves and maps derived from the model show the partial dependency plots (*21*); maps for both data and model are colored from zero (blue) to maximal firing rate (red; value indicated). (**D** and **E**) Two more examples of allocentric cells showing the maps of the top three most important variables (from left to right) and the Shapley values for all of the allocentric variables (rightmost bar graph). Note that in (E), the top three variables all encoded the positions of other bats in absolute allocentric space (*6*). (**F**) Example of an egocentric cell, plotted as in (C). The top three variables encoded by this neuron were the direction ("dir") to one bat, the direction to a second bat, and the distance ("dist") to a third bat. (**G** and **H**) Two more examples of egocentric cells showing the tunings for the top three most important variables (from left to right) and the Shapley values of all the egocentric variables (rightmost). See Fig. 6 for population analyses.

high Shapley values (high contribution) for only one or two variables and low Shapley values for all other variables [Fig. 6B, sparse-coding cells are plotted in the top rows of the two matrices and have high contribution (high Shapley value, yellow) of only one or two behavioral variables and low contribution of the other variables (blue)]. Other cells had more distributed representation, jointly encoding many behavioral variables [Fig. 6B, bottom rows of the two matrices; these neurons have more uniform contribution of all the variables (medium Shapley value, green or blue for all variables); see also Fig. 6C].

Not all behavioral variables were uniformly represented. Consistent with the known tuning of dorsal CA1 neurons to the animal's self place and self head direction (HD) (*15*, *36*, *38*, *39*), we found that a substantial fraction of the allocentric cells showed high contribution of these two self-variables [Fig. 6B, allocentric cells matrix; note the large Shapley values (yellow) in the "place" and "HD" columns; see also fig. S12B]. This self-tuning was most evident in the

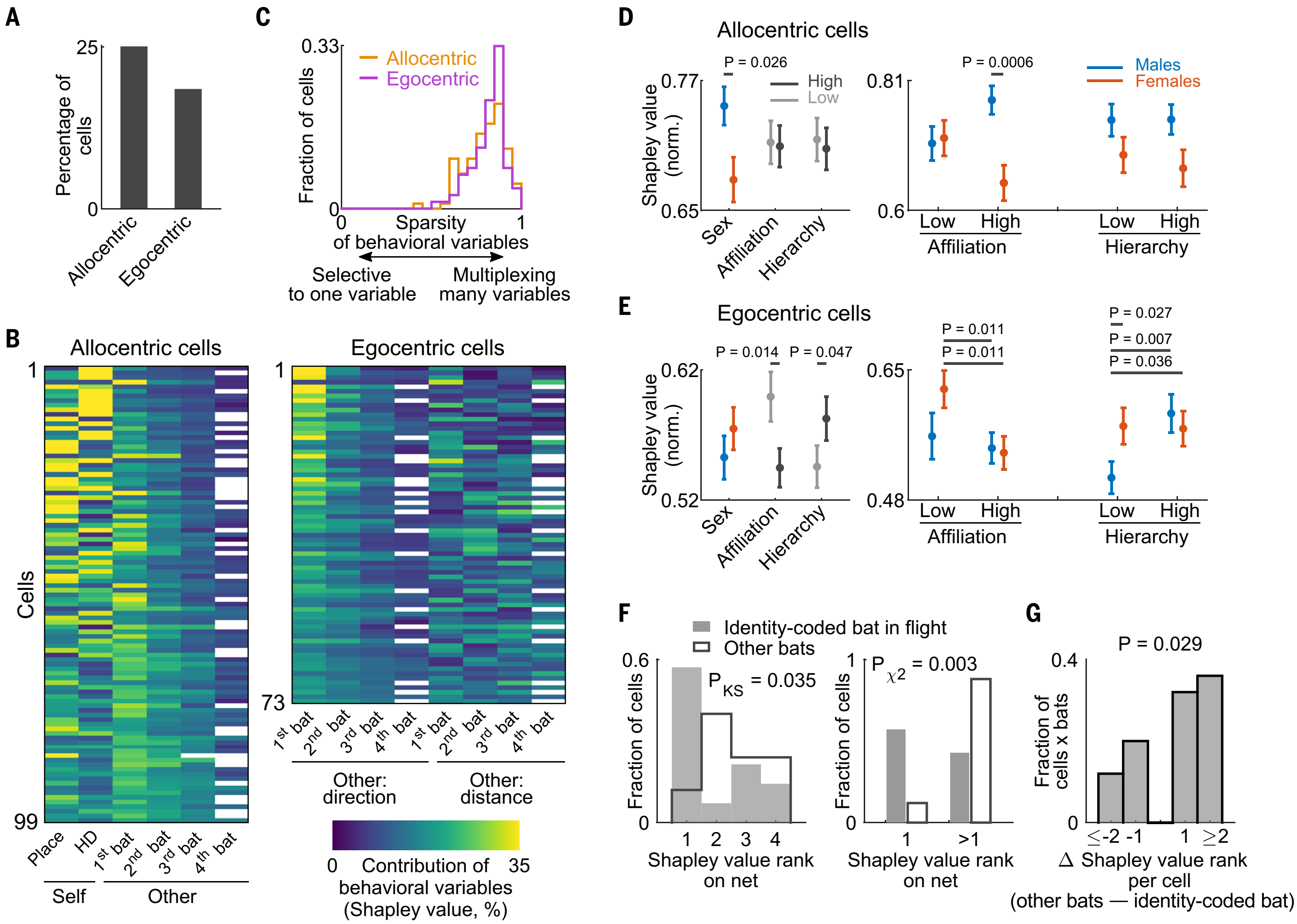

Fig. 6. Hippocampal neurons encode socially important features such as sex, social hierarchy, and social affiliation of other individuals. (**A**) A higher fraction of cells encoded allocentric information (25%, n = 99/394 cells) than egocentric information (19%, n = 73/394 cells); 394 is the number of valid cells that met criteria of sufficient amount of data, with multiple conspecifics being present together on the large net (*21*). (**B**) Population summary for the significant allocentric cells (left) and egocentric cells (right) showing for each of the neurons (rows) the relative contributions (Shapley values) for all of the behavioral variables (columns). Color-coded from blue (zero contribution) to yellow (high contribution; we clipped here the Shapley values at 35% for visualization only). The cells are arranged from top to bottom by the sparsity of the contribution of the behavioral variables, with cells encoding only a few variables plotted at the top and cells with distributed encoding of many variables plotted at the bottom. (**C**) Histograms showing sparsity of behavioral variables of allocentric cells (orange) and egocentric cells (purple). High sparsity indicates multiplexing of many variables, and low sparsity indicates selectivity to one variable (*21*). (**D**) Shapley values for the allocentric cells (normalized by the maximal value in each cell; error bars indicate mean ± SEM). Left, grouping the data by the other bat's sex, social affiliation, or relative social hierarchy (P value indicates main term of a three-way ANOVA). Right, separating the data three-way according to sex (red versus blue), affiliation, and hierarchy of the other bat (P value is after correction for multiple comparisons by Fisher's least-significant difference). "Low" or "high" hierarchy indicates whether the other bat was lower or higher in hierarchy than the recorded bat, respectively, and "low" or "high" affiliation is according to thresholding of the affiliation index (*21*). (**E**) Same as (D) but for egocentric cells. Allocentric cells showed a significant modulation by sex, whereas egocentric cells showed a significant modulation by affiliation and hierarchy (see also table S2). (**F**) Left, histograms of Shapley value rank on the net for the identity-coded bat in-flight (gray bars) and for the other bats that were not identity coded in-flight (white bars). The identity-coded bats in flight had a significantly different distribution of Shapley ranks on the net than the noncoded bats (white bar graph is right shifted compared with the gray; Kolmogorov-Smirnov test, P = 0.035; shown are n = 14 out of the 34 identity-coding cells in Fig. 3 that were also valid for analysis in this figure). Right, most identity-coded bats in-flight (gray bars) had Shapley rank 1 on the net, whereas the noncoded bats in-flight (white bars) had Shapley ranks >1 (χ^2 test: P = 0.003). (**G**) Distribution of the fraction of cells × bats that showed a particular difference in Shapley rank between the noncoded bats in-flight and the identity-coded bat in-flight. Note that many of the cells showed a prominent difference of ≥2 in the Shapley ranks (Wilcoxon signed-rank test, right-tailed, P = 0.029, n = 25 cells × bats).

sparse neurons (the high Shapley values were concentrated mostly in the "place" and "HD" columns at the top part of the matrix): These neurons are classical place cells. However, many cells were not pure place cells but exhibited strong identity coding for one or two other bats (Fig. 6B: Note the yellow color for other bats in the two matrices). Egocentric cells more prominently encoded the directions to other bats compared with the distances to them (Fig. 6B, right matrix: higher Shapley values in the direction columns compared with the distance columns; see also fig. S12C).

To validate that the social encoding specifically represented bats and was not related to objects, rewards, or prominent locations, we controlled for four alternative locations: (i) the most prominent object (a camera suspended from the center of the ceiling, Fig. 1A); (ii) the nearest food bowl, which served as a primary reward location; (iii) the corner of the net closest to where the bats usually flew from, which is a prominent geometrical feature near the bats; and (iv) the median location of each of the bats in the session. We evaluated the GAM egocentric model by replacing each of the four other bats one at a time by each of these alternative locations (*21*) and found a significant reduction in the performance of the GAM model for all. The prediction of the model was significantly lower for all these four control cases [Wilcoxon signed-rank test for comparing the correlations of the model with the data (Spearman ρ) for the original data versus the correlations for each of the four control cases: $P = 1.71 \times 10^{-4}$, $P = 6.68 \times 10^{-5}$, $P = 6.21 \times 10^{-7}$, $P = 5.39 \times 10^{-5}$, respectively; $P = 3.22 \times 10^{-17}$ when pooling the four controls]. Also, the Shapley values for the four control cases were significantly lower than for the real bats (Wilcoxon rank-sum test: $P = 3.70 \times 10^{-4}$, $P = 8.80 \times 10^{-4}$, $P = 6.19 \times 10^{-6}$, $P = 2.11 \times 10^{-2}$, respectively; $P = 6.63 \times 10^{-12}$ when pooling the four controls; similarly, for the Shapley ranks, we found that the four control cases ranked worse than the data: Wilcoxon rank-sum test: $P = 5.38 \times 10^{-3}$, $P = 4.45 \times 10^{-4}$, $P = 1.38 \times 10^{-2}$, $P = 2.83 \times 10^{-2}$, respectively; $P = 4.31 \times 10^{-8}$ when pooling the four controls). This suggests that the real bats were represented more strongly than prominent objects or reward locations.

Representation of sex, dominance hierarchy, and social affiliation

Allocentric and egocentric cells showed substantial differences in how they represented socially relevant features of the conspecifics, including the sex of the other bat, its relative social hierarchy, and social affiliation, namely how affiliated or "friendly" the other bat was to the recorded bat (Fig. 6, D and E, and fig. S13) (*21*). These three social features, sex, hierarchy, and affiliation, were statistically independent of each other (*21*). We found that allocentric cells were significantly modulated by sex, with males being represented more strongly than females (Fig. 6D, left panel; three-way ANOVA: $P = 0.026$ for the sex term), which is akin to the stronger encoding of males during flights (Fig. 3H). Again, this could not be explained by whether the sex of the other bat was the same or different from the recorded bat (three-way ANOVA: $P = 0.84$). In particular, high-affiliation male conspecifics were represented more strongly than high-affiliation female conspecifics [Fig. 6D, right panel; $P = 0.0006$, corrected for multiple comparisons (Fisher's least-significant difference); see table S2 for full ANOVA results; similar results were obtained when using a mixed-effect linear model (table S2)]. By contrast, egocentric cells were significantly modulated by affiliation and by relative hierarchy, exhibiting stronger coding of low-affiliation bats (less-friendly animals) and of high-hierarchy bats (more-dominant animals) (Fig. 6E, left panel; three-way ANOVA: $P = 0.014$ for the affiliation term, $P = 0.047$ for the hierarchy term). However, the effect was more complex when taking sex into account. The feature that most strongly affected the neural responses to male conspecifics was hierarchy, whereas for female conspecifics it was affiliation. In particular, lower-hierarchy males were represented more weakly than higher-hierarchy males and also more weakly than both lower- and higher-hierarchy females (Fig. 6E, right panel; $P = 0.007$, $P = 0.027$, and $P = 0.036$, respectively, corrected for multiple comparisons; see table S2 for results of ANOVA and mixed-effect linear model). Lower-affiliation females were represented more strongly than both higher-affiliation females and higher-affiliation males (Fig. 6E, right panel; $P = 0.011$ and $P = 0.011$, respectively, corrected for multiple comparisons). Overall, bat hippocampal neurons represented important social features of conspecifics, such as their position, direction, and identity, and were modulated by their sex, social hierarchy, and social affiliation.

Identity coding by hippocampal neurons is context invariant

Next, we investigated whether identity coding by hippocampal neurons is independent of context, with the same bats being represented in different locations and across different behavioral states. To this end, we examined whether the cells that represented the identities of specific bats in-flight (Fig. 3B) also represented the same bats during active observation on-net (Fig. 6B). We found that for most cells, the identity-coded bat in-flight was also the most strongly coded bat on the net, namely the bat with the highest Shapley value among the conspecific bats (Fig. 6F). Conversely, all of the other bats, which were not identity coded in-flight, were also more weakly coded on the net by the cells (Fig. 6F, left panel: Kolmogorov-Smirnov test, $P = 0.035$; right panel: χ^2 test, comparing if the identity-coded bat in-flight had a Shapley rank 1 on-net or not versus the lower-ranked bats: $P = 0.003$). Further, the difference in ranks between the identity-coded and noncoded bats was substantial, often ≥2 (Fig. 6G). Together, these analyses indicate that a given hippocampal neuron encoded the identity of the same bat across different contexts, i.e., across different locations (midroom versus net) and across different behavioral states (flight versus active-on-net)—suggesting an invariant identity coding.

Finally, we investigated which brain regions receive these hippocampal signals encoding social identity, sex, hierarchy, affiliation, and interactions. We examined the anatomical projections from dorsal CA1 in the bat by injecting anterograde tracers into the dorsal CA1 of four additional bats (*21*). We found that the dorsal CA1 sends direct monosynaptic (albeit sparse) projections to multiple brain areas, including the medial prefrontal infralimbic cortex, anterior cingulate cortex, frontal cortex, orbitofrontal cortex, substantia nigra, nucleus accumbens, retrosplenial cortex, and claustrum (fig. S14)—regions that are involved in regulating social behaviors (*40*). We hypothesize that the rich social signals that are represented in the hippocampus during natural behaviors may shape social neural information throughout the brain.

Discussion

We found here that many hippocampal CA1 neurons represented the identity of conspecifics and encoded their social features, such as their sex, dominance hierarchy, and social affiliation, when animals were in a naturalistic social group and allowed to perform unconstrained, untrained, natural social behaviors. These representations were sociospatial in nature (*5*, *6*, *41*), binding together the well-known hippocampal spatial representations (*14*–*16*) with the more social conditions that group-living animals typically encounter in the natural world. We found that during flight, hippocampal spatial coding was strongly modulated by social context through rate remapping, both prospectively and retrospectively, and exhibited strong identity coding (Figs. 2 and 3). Social coding of individual bats exhibited invariance between the two flight directions (Fig. 3, F and G). During social-interaction events, many hippocampal neurons responded selectively to particular types of social interactions (Fig. 4). When the bats were active on the net, hippocampal neurons combined representations of self and other: Cells could encode others' locations in allocentric or egocentric space and could exhibit identity coding, representing one or two individuals or distributed encoding of many individuals (Fig. 6, B and C). The finding of identity coding by hippocampal neurons prompted us to investigate which social features of individuals were being represented, and we found that these neurons represented sex, social hierarchy, and social affiliation (Fig. 6, D and E), all highly important social features. Finally, we found invariance of social representations across behavioral contexts, both in-flight and on-net: Neural encoding of identities was invariant (Fig. 6, F and G) and males were represented more strongly than females (Figs. 3H and 6D) across these contexts. Thus, the hippocampus carries strong and rich social signals when studied in a social group.

Our findings of strong social representations in hippocampal dorsal CA1 are markedly different from previous studies finding weak social responses in that area (8, 42, 43). This difference is possibly due to the naturalistic social setting in our study, in which the animals were free to decide what to do, when to do it, and with which other bats to engage socially. The previous experiments typically involved artificial social tasks imposed by the experimenters. This suggests that the hippocampus represents behavioral variables only if they are relevant. Consistent with this view, whereas here we found strong identity coding, this was not the case in a previous study (44), where we found that during high-speed collision avoidance between two bats, hippocampal CA1 neurons did not exhibit identity coding, most likely because when avoiding collisions, it is less relevant to encode who you are colliding with. Similarly, we found here that CA1 neurons encoded directions to other bats more strongly than distances to them (Fig. 6B, right, and fig. S12C), possibly because in a very small environment (i.e., the nets), directions are more relevant than distances. Thus, hippocampal neurons seem to represent only relevant social information.

The finding of social encoding during flight cannot be explained as a general gating by internal states such as surprise, arousal, fear, or reward, for the following reasons. Social responses occurred at all possible locations in-flight (Fig. 2F), which is incompatible with a general response to surprise, arousal, fear, or reward, all of which should be focused near the locations of social encounters at the nets. Also, some cells were socially modulated after takeoff from a net (Fig. 2D), i.e., when going away from a social agent, which is also incompatible with surprise, arousal, or reward. Some cells responded more strongly for an empty net (Fig. 2, A to D, "alone"), which is inconsistent with a social reward near other bats or fear response to other bats. Moreover, in simultaneously recorded neurons, one cell could respond more strongly to the social condition and another to the alone condition (e.g., Fig. 2B: cells 71 and 70, respectively), which is inconsistent with a general gating response by an internal state. The fraction of socially modulated cells (Fig. 2E) was >10-fold larger than the previously reported fraction of CA1 cells encoding nonsocial rewards (45), indicating that the social responses that we observed could not be explained by cells encoding nonsocial rewards. Many cells showed social identity coding, representing specific bats (Figs. 3, B and D, and 6F), a specificity that is inconsistent with a general gating response by internal states (46). Finally, many more cells responded to social interactions than to other salient nonsocial events such as landing and takeoff of other bats near the recorded bat (fig. S8C), indicating that surprise or arousal were unlikely to account for these responses. Together, these findings suggest that the social responses that we observed are not explained by a general response to internal states.

Our findings are distinct from previous reports of social place cells (5, 6), because here we showed hippocampal identity coding in a multianimal group. Our results are also distinct from a previous report on population-level identity coding in mouse hippocampus (47), because the task was very artificial and the identity coding was largely driven by training. Further, a recent study (13) investigated hippocampal activity in groups of bats flying in a room and showed identity coding only in-flight. Consistent with our results, the latter study showed that reward cannot explain social coding in-flight. However, our naturalistic setup allowed us to also investigate social interactions and to study the representations of sex, dominance hierarchy, and social affiliation, none of which was investigated in previous hippocampal studies (13, 47). In addition, we were able to reveal multiple invariances in social coding, and discover that in a naturalistic group setting, hippocampal neurons represent highly multidimensional social information.

We found that hippocampal neurons responded to social interaction events, and disambiguated between interaction types (Fig. 4). Further investigation would be required to determine whether hippocampal neurons respond to specific motor patterns that occur during social interactions or to the abstract notion of a social interaction. Cells that encode social interactions have been previously observed in the dorsomedial prefrontal cortex (48, 49) and in the anterior cingulate cortex (50). However, to our knowledge, our finding is the first in the hippocampus, a brain area that is crucial for encoding memories for events, including events of social interactions (51, 52).

We have shown here that hippocampal CA1 neurons encode several social features that are relevant in natural social settings, specifically, sex, social hierarchy, and social affiliation. These features were previously studied separately (10, 26, 53, 54), but here we showed that they are all encoded simultaneously in the same neuronal population (Fig. 6, D and E). Thus, single hippocampal neurons can multiplex information that tracks the multidimensional social information in natural social settings [see also (55)]. Allocentric and egocentric cells represented these social features differently, with allocentric neurons encoding sex and egocentric neurons encoding affiliation and hierarchy (Fig. 6, D and E). This may suggest that the allocentric and egocentric cell populations receive different inputs. The social representations that we observed were typically conjoined with spatial representations (Figs. 2 to 6), perhaps indicating that the spatial cognitive map system also evolved to include the coding of social information into a single cognitive map–like framework: a sociospatial cognitive map (6, 36, 54).

Finally, we found that a large fraction of hippocampal cells were strongly modulated by social identity, both in-flight (Fig. 3) and on-net (Fig. 6). The identity coding was conserved across different locations and contexts: An identity-coding neuron typically represented the same bat both in-flight and on-net (Fig. 6F) and in both flight directions (Fig. 3, F and G). This identity coding was likely based on multisensory information (56), including visual, auditory, and olfactory cues, but additional studies would be required to assess the contribution of each sensory modality to the encoding of individual identity and to the coding of social features such as sex, social hierarchy, and social affiliation. The identity coding that we found here is reminiscent of previous reports on identity-coding neurons in the hippocampal formation of humans and monkeys that passively viewed images on a screen (12, 56), but here these were uncovered during active social interactions. The fraction of socially responsive hippocampal neurons in our study was much higher than in the passively viewing humans or monkeys in previous studies. We speculate that our finding of a very high fraction of socially coding cells in the bat hippocampus may be a consequence of three factors: (i) our use of a mixed-sex group, (ii) our highly naturalistic setup and natural social interactions, and (iii) the fact that these were wild bats that have experienced the full natural repertoire of social interactions outdoors. We therefore suggest that future investigations should include brain studies in mixed-sex social groups, in which animals can perform natural social interactions—that are among the most important behaviors that animals have evolved.

Materials and methods summary

We conducted tetrode-based recordings of single neurons in the dorsal hippocampus area CA1 of wild Egyptian fruit bats (*R. aegyptiacus*) of both sexes. The bats lived continuously in a mixed-sex social colony of five to 10 individuals and behaved freely without any tasks. We tracked the positions and identities of all bats using a combination of a radiofrequency-based system and a video-based system, and we video-documented the bats' social interactions. We computed social and spatial responses of individual neurons when the bats were flying, engaged in social interactions, or were active on the landing nets. Further details can be found in the supplementary materials and methods (21).

REFERENCES AND NOTES

1. T. D. Seeley, *The Wisdom of the Hive: The Social Physiology of Honey Bee Colonies* (Harvard Univ. Press, 2009). doi: 10.2307/j.ctv1kz4h15
2. A. Strandburg-Peshkin, D. R. Farine, I. D. Couzin, M. C. Crofoot, Shared decision-making drives collective movement in wild

baboons. *Science* **348**, 1358–1361 (2015). doi: 10.1126/science.aaa5099; pmid: 26089514
3. T. Clutton-Brock, *Mammal Societies* (Wiley, 2016).
4. T. Okuyama, T. Kitamura, D. S. Roy, S. Itohara, S. Tonegawa, Ventral CA1 neurons store social memory. *Science* **353**, 1536–1541 (2016). doi: 10.1126/science.aaf7003; pmid: 27708103
5. T. Danjo, T. Toyoizumi, S. Fujisawa, Spatial representations of self and other in the hippocampus. *Science* **359**, 213–218 (2018). doi: 10.1126/science.aao3898; pmid: 29326273
6. D. B. Omer, S. R. Maimon, L. Las, N. Ulanovsky, Social place-cells in the bat hippocampus. *Science* **359**, 218–224 (2018). doi: 10.1126/science.aao3474; pmid: 29326274
7. R. P. Rao, M. von Heimendahl, V. Bahr, M. Brecht, Neuronal responses to conspecifics in the ventral CA1. *Cell Rep.* **27**, 3460–3472.e3 (2019). doi: 10.1016/j.celrep.2019.05.081; pmid: 31216468
8. M. von Heimendahl, R. P. Rao, M. Brecht, Weak and nondiscriminative responses to conspecifics in the rat hippocampus. *J. Neurosci.* **32**, 2129–2141 (2012). doi: 10.1523/JNEUROSCI.3812-11.2012; pmid: 22323725
9. T. Zhou *et al*., History of winning remodels thalamo-PFC circuit to reinforce social dominance. *Science* **357**, 162–168 (2017). doi: 10.1126/science.aak9726; pmid: 28706064
10. Y. Li *et al*., Neuronal representation of social information in the medial amygdala of awake behaving mice. *Cell* **171**, 1176–1190.e17 (2017). doi: 10.1016/j.cell.2017.10.015; pmid: 29107332
11. F. L. Hitti, S. A. Siegelbaum, The hippocampal CA2 region is essential for social memory. *Nature* **508**, 88–92 (2014). doi: 10.1038/nature13028; pmid: 24572357
12. R. Q. Quiroga, L. Reddy, G. Kreiman, C. Koch, I. Fried, Invariant visual representation by single neurons in the human brain. *Nature* **435**, 1102–1107 (2005). doi: 10.1038/nature03687; pmid: 15973409
13. A. Forli, M. M. Yartsev, Hippocampal representation during collective spatial behaviour in bats. *Nature* **621**, 796–803 (2023). doi: 10.1038/s41586-023-06478-7; pmid: 37648869
14. J. O'Keefe, J. Dostrovsky, The hippocampus as a spatial map. Preliminary evidence from unit activity in the freely-moving rat. *Brain Res.* **34**, 171–175 (1971). doi: 10.1016/0006-8993(71)90358-1; pmid: 5124915
15. M. A. Wilson, B. L. McNaughton, Dynamics of the hippocampal ensemble code for space. *Science* **261**, 1055–1058 (1993). doi: 10.1126/science.8351520; pmid: 8351520
16. N. Ulanovsky, C. F. Moss, Hippocampal cellular and network activity in freely moving echolocating bats. *Nat. Neurosci.* **10**, 224–233 (2007). doi: 10.1038/nn1829; pmid: 17220886
17. G. G. Kwiecinski, T. A. Griffiths, Rousettus egyptiacus. *Mamm. Species* **611**, 1–9 (1999). doi: 10.2307/3504411
18. G. Neuweiler, *The Biology of Bats* (Oxford Univ. Press, 2000).
19. M. C. Rose, B. Styr, T. A. Schmid, J. E. Elie, M. M. Yartsev, Cortical representation of group social communication in bats. *Science* **374**, eaba9584 (2021). doi: 10.1126/science.aba9584; pmid: 34672724
20. L. Las, N. Ulanovsky, Bats. *Nat. Methods* **21**, 1135–1137 (2024). doi: 10.1038/s41592-024-02330-6; pmid: 38997589
21. Materials and methods are available as supplementary materials.
22. J. D. Crall, N. Gravish, A. M. Mountcastle, S. A. Combes, BEEtag: A low-cost, image-based tracking system for the study of animal behavior and locomotion. *PLOS ONE* **10**, e0136487 (2015). doi: 10.1371/journal.pone.0136487; pmid: 26332211
23. R. A. Hinde, Interactions, relationships and social structure. *Man (Lond.)* **11**, 1–17 (1976). doi: 10.2307/2800384
24. S. Shultz, C. Opie, Q. D. Atkinson, Stepwise evolution of stable sociality in primates. *Nature* **479**, 219–222 (2011). doi: 10.1038/nature10601; pmid: 22071768
25. S. W. Li *et al*., Frontal neurons driving competitive behaviour and ecology of social groups. *Nature* **603**, 661–666 (2022). doi: 10.1038/s41586-021-04000-5; pmid: 35296863
26. F. Wang *et al*., Bidirectional control of social hierarchy by synaptic efficacy in medial prefrontal cortex. *Science* **334**, 693–697 (2011). doi: 10.1126/science.1209951; pmid: 21960531
27. B. L. McNaughton, C. A. Barnes, J. O'Keefe, The contributions of position, direction, and velocity to single unit activity in the hippocampus of freely-moving rats. *Exp. Brain Res.* **52**, 41–49 (1983). doi: 10.1007/BF00237147; pmid: 6628596
28. M. Geva-Sagiv, S. Romani, L. Las, N. Ulanovsky, Hippocampal global remapping for different sensory modalities in flying bats. *Nat. Neurosci.* **19**, 952–958 (2016). doi: 10.1038/nn.4310; pmid: 27239936
29. W. A. Liberti 3rd, T. A. Schmid, A. Forli, M. Snyder, M. M. Yartsev, A stable hippocampal code in freely flying bats. *Nature* **604**, 98–103 (2022). doi: 10.1038/s41586-022-04560-0; pmid: 35355012
30. T. Eliav *et al*., Multiscale representation of very large environments in the hippocampus of flying bats. *Science* **372**, eabg4020 (2021). doi: 10.1126/science.abg4020; pmid: 34045327
31. K. W. Latimer, J. L. Yates, M. L. Meister, A. C. Huk, J. W. Pillow, Single-trial spike trains in parietal cortex reveal discrete steps during decision-making. *Science* **349**, 184–187 (2015). doi: 10.1126/science.aaa4056; pmid: 26160947
32. K. Hardcastle, N. Maheswaranathan, S. Ganguli, L. M. Giocomo, A multiplexed, heterogeneous, and adaptive code for navigation in medial entorhinal cortex. *Neuron* **94**, 375–387.e7 (2017). doi: 10.1016/j.neuron.2017.03.025; pmid: 28392071
33. J.-P. Noel *et al*., Coding of latent variables in sensory, parietal, and frontal cortices during closed-loop virtual navigation. *eLife* **11**, e80280 (2022). doi: 10.7554/eLife.80280; pmid: 36282071
34. N. L. Goodwin, S. R. O. Nilsson, J. J. Choong, S. A. Golden, Toward the explainability, transparency, and universality of machine learning for behavioral classification in neuroscience. *Curr. Opin. Neurobiol.* **73**, 102544 (2022). doi: 10.1016/j.conb.2022.102544; pmid: 35487088
35. L. S. Shapley, "A value for n-person games" in *Contributions to the Theory of Games*, H. W. Kuhn, A. W. Tucker, Eds. (Princeton Univ. Press, 1953), vol. II, pp. 307–318.
36. J. O'Keefe, L. Nadel, *The Hippocampus as a Cognitive Map* (Oxford Univ. Press, 1978).
37. A. Sarel, A. Finkelstein, L. Las, N. Ulanovsky, Vectorial representation of spatial goals in the hippocampus of bats. *Science* **355**, 176–180 (2017). doi: 10.1126/science.aak9589; pmid: 28082589
38. L. Acharya, Z. M. Aghajan, C. Vuong, J. J. Moore, M. R. Mehta, Causal influence of visual cues on hippocampal directional selectivity. *Cell* **164**, 197–207 (2016). doi: 10.1016/j.cell.2015.12.015; pmid: 26709045
39. A. Rubin, M. M. Yartsev, N. Ulanovsky, Encoding of head direction by hippocampal place cells in bats. *J. Neurosci.* **34**, 1067–1080 (2014). doi: 10.1523/JNEUROSCI.5393-12.2014; pmid: 24431464
40. R. Adolphs, The social brain: Neural basis of social knowledge. *Annu. Rev. Psychol.* **60**, 693–716 (2009). doi: 10.1146/annurev.psych.60.110707.163514; pmid: 18771388
41. M. Murugan *et al*., Combined social and spatial coding in a descending projection from the prefrontal cortex. *Cell* **171**, 1663–1677.e16 (2017). doi: 10.1016/j.cell.2017.11.002; pmid: 29224779
42. X. Mou, D. Ji, Social observation enhances cross-environment activation of hippocampal place cell patterns. *eLife* **5**, e18022 (2016). doi: 10.7554/eLife.18022; pmid: 27692067
43. L. Zynyuk, J. Huxter, R. U. Muller, S. E. Fox, The presence of a second rat has only subtle effects on the location-specific firing of hippocampal place cells. *Hippocampus* **22**, 1405–1416 (2012). doi: 10.1002/hipo.20977; pmid: 21997883
44. A. Sarel *et al*., Natural switches in behaviour rapidly modulate hippocampal coding. *Nature* **609**, 119–127 (2022). doi: 10.1038/s41586-022-05112-2; pmid: 36002570
45. J. L. Gauthier, D. W. Tank, A dedicated population for reward coding in the hippocampus. *Neuron* **99**, 179–193.e7 (2018). doi: 10.1016/j.neuron.2018.06.008; pmid: 30008297
46. M. A. Moita, S. Rosis, Y. Zhou, J. E. LeDoux, H. T. Blair, Hippocampal place cells acquire location-specific responses to the conditioned stimulus during auditory fear conditioning. *Neuron* **37**, 485–497 (2003). doi: 10.1016/S0896-6273(03)00033-3; pmid: 12575955
47. E. Kong, K.-H. Lee, J. Do, P. Kim, D. Lee, Dynamic and stable hippocampal representations of social identity and reward expectation support associative social memory in male mice. *Nat. Commun.* **14**, 2597 (2023). doi: 10.1038/s41467-023-38338-3; pmid: 37147388
48. R. Báez-Mendoza, E. P. Mastrobattista, A. J. Wang, Z. M. Williams, Social agent identity cells in the prefrontal cortex of interacting groups of primates. *Science* **374**, eabb4149 (2021). doi: 10.1126/science.abb4149; pmid: 34672743
49. L. Kingsbury *et al*., Correlated neural activity and encoding of behavior across brains of socially interacting animals. *Cell* **178**, 429–446.e16 (2019). doi: 10.1016/j.cell.2019.05.022; pmid: 31230711
50. K. Haroush, Z. M. Williams, Neuronal prediction of opponent's behavior during cooperative social interchange in primates. *Cell* **160**, 1233–1245 (2015). doi: 10.1016/j.cell.2015.01.045; pmid: 25728667
51. N. Burgess, E. A. Maguire, J. O'Keefe, The human hippocampus and spatial and episodic memory. *Neuron* **35**, 625–641 (2002). doi: 10.1016/S0896-6273(02)00830-9; pmid: 12194864
52. W. B. Scoville, B. Milner, Loss of recent memory after bilateral hippocampal lesions. *J. Neurol. Neurosurg. Psychiatry* **20**, 11–21 (1957). doi: 10.1136/jnnp.20.1.11; pmid: 13406589
53. L. Kingsbury *et al*., Cortical representations of conspecific sex shape social behavior. *Neuron* **107**, 941–953.e7 (2020). doi: 10.1016/j.neuron.2020.06.020; pmid: 32663438
54. R. M. Tavares *et al*., A map for social navigation in the human brain. *Neuron* **87**, 231–243 (2015). doi: 10.1016/j.neuron.2015.06.011; pmid: 26139376
55. E. H. Nieh *et al*., Geometry of abstract learned knowledge in the hippocampus. *Nature* **595**, 80–84 (2021). doi: 10.1038/s41586-021-03652-7; pmid: 34135512
56. T. J. Tyree, M. Metke, C. T. Miller, Cross-modal representation of identity in the primate hippocampus. *Science* **382**, 417–423 (2023). doi: 10.1126/science.adf0460; pmid: 37883535
57. S. Ray *et al*., Data for: Hippocampal coding of identity, sex, hierarchy and affiliation in a social group of wild fruit bats, Zenodo (2025); https://doi.org/10.5281/zenodo.13938638.

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SUPPLEMENTARY MATERIALS

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Materials and Methods
Figs. S1 to S14
Tables S1 and S2
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MDAR Reproducibility Checklist

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